2014 D

SOLID PHASE CONDUCTION

CHARGE

$$\sigma_i = n_i q_i \mu_i$$

MOBILITY

DEFECT 'i'

CONCENTRATION CHARGE CARRIER

KRÖGER-VINK NOTATION

DEFECT $\longrightarrow$ $Fe_{Mg}^{\bullet}$

CHARGE RELATIVE TO SITE

SITE

CHARGE	
-3	'''
-2	''
-1	'
0	x
1	•
2	••
3	•••

$$n_i = N_i \exp\left(-\frac{\Delta H_{n_i}}{k_B T}\right)$$

ACTIVATION ENTHALPY

TEMP. IN K

MAXIMUM CONCENTRATION

BOLTZMANN'S CONSTANT

$$\mu_i = \mu_{0,i} \exp\left(-\frac{\Delta H_{\mu_i}}{k_B T}\right)$$

$$\sigma_i = N_i \exp\left(-\frac{\Delta H_{n_i}}{k_B T}\right) q_i \mu_{i,0} \exp\left(-\frac{\Delta H_{\mu_i}}{k_B T}\right)$$

$$= \underbrace{(N_i q_i \mu_{i,0})}_{\sigma_{i,0}} \exp\left(-\frac{\overbrace{(\Delta H_{n_i} + \Delta H_{\mu_i})}^{\Delta H_i}}{k_B T}\right)$$

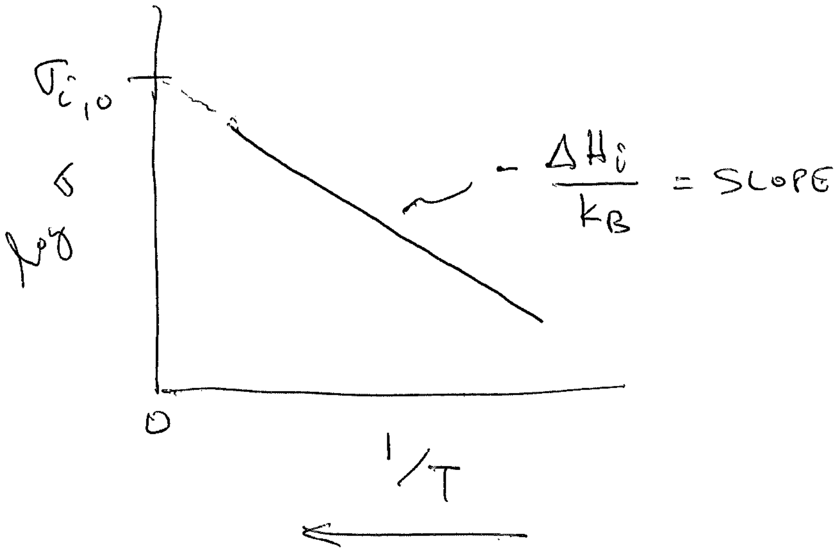
$$\sigma_i = \sigma_{i,0} \exp\left(-\frac{\Delta H_i}{k_B T}\right)$$

$$\log \sigma_i = \log \left(\sigma_{i,0} \exp \left(- \frac{\Delta H_i}{k_B T} \right) \right)$$

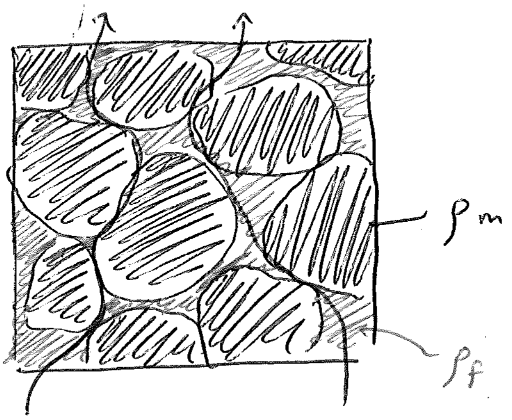
$$= \log \sigma_{i,0} - \frac{\Delta H_i}{k_B T}$$

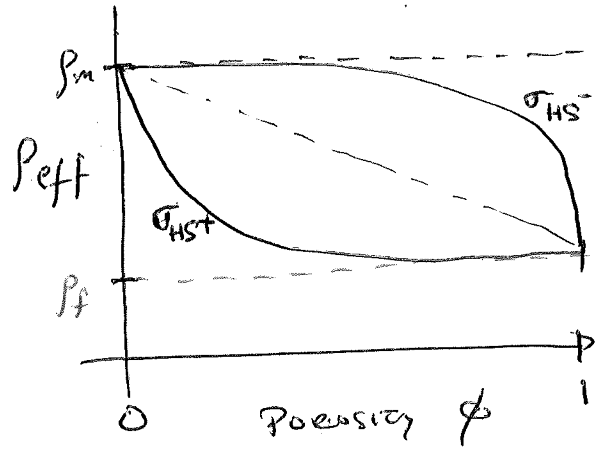
NATURAL LOG

$$y(T) = b + m \frac{1}{T} = \text{LINE} \quad \frac{1}{T}$$



CONDUCTIVITY





SOLID PHASE CONDUCTION:

$$\underset{\substack{\uparrow \\ \text{DEFECT "i"}}}{\sigma_i} = \underset{\substack{\uparrow \\ \text{CONCENTRATION}}}{n_i} \overset{\text{CHARGE}}{q_i} \underset{\text{MOBILITY}}{\mu_i}$$

MOBILITY

$$\mu_i = \mu_{o,i} \exp\left(- \frac{\overset{\text{ACTIVATION ENTHALPY}}{\Delta H_{\mu i}}}{k_B T}\right)$$

k_B: BOLTZMANN'S CONSTANT

T: TEMPERATURE IN [K]

$$n_i = N_i \exp\left(- \frac{\Delta H_{n i}}{k_B T}\right)$$

$$\sigma_i = N_i \exp\left(- \frac{\Delta H_{n i}}{k_B T}\right) q_i \mu_{o,i} \exp\left(- \frac{\Delta H_{\mu i}}{k_B T}\right)$$

$$= \underbrace{N_i q_i \mu_{o,i}}_{\substack{\sigma_{i,o} \\ \text{PRE-MULTIPLIER}}} \exp\left(- \frac{\overbrace{(\Delta H_{n i} + \Delta H_{\mu i})}^{\Delta H_i}}{k_B T}\right)$$

$$\sigma_i = \sigma_{i,o} \exp\left(- \frac{\Delta H_i}{k_B T}\right)$$

$$\log \sigma_i = \log\left(\sigma_{i,0} \exp\left(-\frac{\Delta H_i}{k_B T}\right)\right)$$

$$= \log \sigma_{i,0} + \log\left(\exp\left(-\frac{\Delta H_i}{k_B T}\right)\right)$$

$$\boxed{\log \sigma_i = \log \sigma_{i,0} - \frac{\Delta H_i}{k_B T}}$$

log = natural log

~~ARRHE~~ ARRHENIUS RELATIONSHIP

$\log \sigma_i = y$ $\quad \log \sigma_{i,0} = b$ $\quad -\frac{\Delta H_i}{k_B} = m$ $\quad \frac{1}{T} = x$

$$y = b + mx$$

$$\sigma_{eff} = \sum_{i=1}^{N} \sigma_i$$

MAXWELL'S EQUATIONS

(1) $\nabla \cdot \vec{E} = \frac{\rho_f}{\varepsilon_0}$ GAUSS'S LAW ELECTRICITY

(2) $\nabla \cdot \vec{H} = 0$ GAUSS'S LAW OF MAGNETISM

(3) $\nabla \times \vec{E} = -\mu_0 \frac{\partial H}{\partial t}$ FARADAY'S LAW

(4) $\nabla \times \vec{H} = \vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$ AMPERE'S LAW

(5) $\vec{J} = \sigma \vec{E}$ CONSTITUTIVE RELATION

(5) INTO (4) $\nabla \times \vec{H} = \sigma \vec{E} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$

(3) INTO (4) $\nabla \times (\nabla \times \vec{E}) = \nabla \times \left(-\mu_0 \frac{\partial \vec{H}}{\partial t}\right)$ SWAP DERIVATIVE ORDER

$$\nabla \times (\nabla \times \vec{E}) = -\mu_0 \frac{\partial}{\partial t}\left(\sigma \vec{E} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t}\right)$$

VECTOR IDENTITY

$$\nabla \times \nabla \times \vec{A} = \nabla \left(\cancel{\nabla \cdot \vec{A}}\right)^{0} - \nabla^2 \vec{A}$$

FROM (1) $\nabla \cdot \vec{E} = \frac{\rho_f}{\varepsilon_0} = 0$

$$\nabla^2 \vec{E} = \mu_0 \sigma \frac{\partial}{\partial t} \vec{E} + \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t}$$ WAVE EQUATION

QUASISTATIC APPROXIMATION

$\sigma << \omega \varepsilon_0$

ANGULAR FREQUENCY

$$\nabla^2 \vec{E} = \mu_0 \sigma \frac{\partial \vec{E}}{\partial t}$$ DIFFUSION EQUATION

$$\vec{E}(x,y,z,t) \xleftrightarrow{\mathcal{F}} \tilde{E}(x,y,z,\omega)$$

time-domain frequency domain

1-D: $$\frac{\partial^2 \vec{E}}{\partial z^2} = \mu_0 \sigma \frac{\partial \vec{E}}{dt}$$ Time-domain

$$\frac{\partial^2 \tilde{E}}{\partial z^2} = -k_z^2 \tilde{E}$$ Frequency-domain

$$\frac{\partial e^{kz}}{\partial z} = ke^{kz}$$

$$\frac{\partial^2 e^{kz}}{\partial z^2} = k^2 e^{kz}$$ CLOSE ⇒ NEED COMPLEX

SOLN:

$$\tilde{E} = \underbrace{E^{(+)}(k_z,\omega)\exp(+ik_z z)}_{\text{UPGOING}} + \underbrace{E^{(-)}(k_z,\omega)\exp(-ik_z z)}_{\text{DOWN GOING}}$$

$$k_z = (1-i)\sqrt{\frac{\sigma\mu_0\omega}{2}}$$ WAVE NUMBER

$i \equiv \sqrt{-1}$

REAL PART, SKIN DEPTH.

$$\delta = \left(\frac{2}{\sigma\mu_0\omega}\right)^{1/2} = [m]$$

3.

$$\sigma_1 = \frac{1}{\rho_1}$$

$$\sigma_2 = \frac{1}{\rho_2}$$

HIGHLY CONDUCTIVE BASEMENT, $\sigma_2 >> \sigma_1$ $\omega \to \infty$

$$k = (1-i)\left(\frac{\sigma \mu_0 \omega}{2}\right)^{1/2}$$

$$Z(0) \approx \omega \mu_0 \left(\frac{1 + i k_2 h_1}{k_2}\right)$$

$$Z(0) \approx i\omega \mu_0 h_1 \qquad \rho_a \propto h_1^2$$

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4. HIGHLY RESISTIVE BASEMENT $\rho_2 \to \infty$, $\sigma_1 >> \sigma_2$

$|k_1| >> |k_2|$

$$Z(0) \approx \frac{\omega \mu_0}{i k_1^2 h_1} \approx \frac{1}{\sigma_1 h_1}$$

$$\rho_a \propto \frac{1}{(\sigma_1 h_1)^2} \qquad \sigma h \equiv S \equiv \text{conductance}$$

$$\rho_a \propto \frac{1}{S^2}$$

$$E_x = E_0 e^{-ik_z z}$$

$$H_y = H_0 e^{-ik_z z} = \frac{k_z}{\omega \mu_0} E_0 e^{-ikz}$$

$$Z_{xy} = \frac{E_x}{H_y} = \frac{E_0 \cancel{\exp(-ik_z z)}}{H_0 \cancel{\exp(-ik_z z)}}$$

$$= \frac{E_0}{H_0}$$

$$= \frac{\cancel{E_0}}{\frac{k_z}{\omega \mu_0} \cancel{E_0}}$$

$$Z_{xy} = \frac{\omega \mu_0}{k_z}$$ INPUTTING k_z

$$= (1+i)\left(\frac{\omega \mu_0}{2\sigma_a}\right)^{1/2}$$

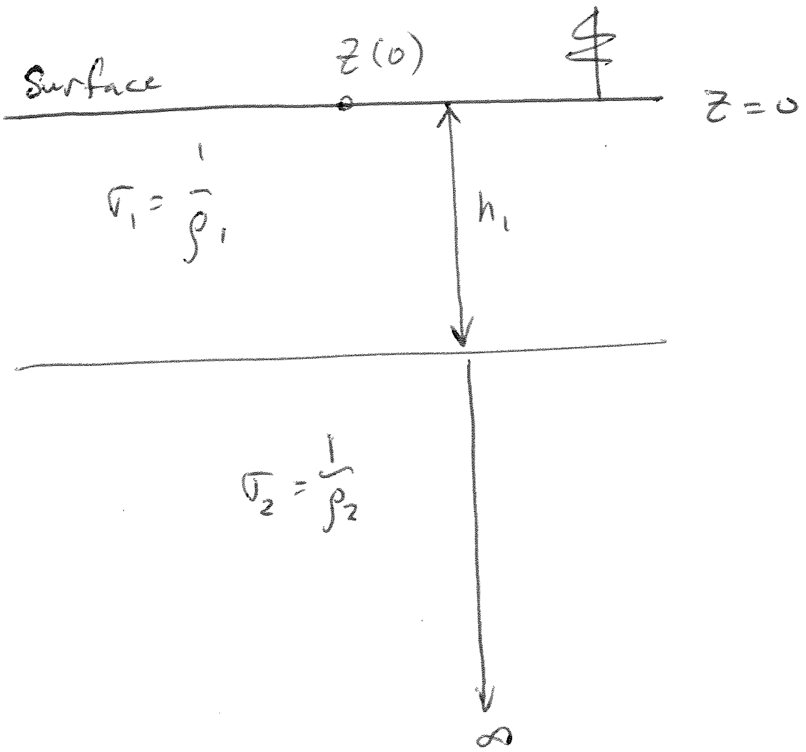
$$\sigma_a = \frac{1}{\rho_a}$$

$$Z_{xy} = (1+i)\left(\frac{\omega \mu_0 \rho_a}{2}\right)^{1/2}$$

$$\rho_a(\omega) = -\frac{1}{\omega \mu_0} Z_{xy}^2$$

$$\rho_a = \frac{1}{\omega \mu_0} |Z_{xy}|^2$$ APPARENT RESISTIVITY

$$Z_{xy} = |Z_{xy}| e^{i\phi}$$ ϕ - PHASE



$$Z(0) = \frac{\omega\mu_0}{k_1}\left(\frac{k_1 + k_2 \tanh(ik_1h_1)}{k_2 + k_1 \tanh(ik_1h_1)}\right)$$

$$\tanh z = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}}$$

$$\tanh(0) = 0$$

$$\lim_{z\to\infty} \tanh(z) = 1$$

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1. THICK SURFACE LAYER, $h_1 \to \infty$ OR $\omega \to \infty$

$$\lim_{h_1\to\infty} \tanh(ik_1h_1) \Rightarrow 1$$

$$Z(0) = \frac{\omega\mu_0}{k_1}\left(\cancelto{1}{\frac{k_1 + k_2}{k_2 + k_1}}\right)$$

$$\boxed{Z(0) = \frac{\omega\mu_0}{k_1} = Z_1 \qquad \rho_a \to \rho_1}$$

2. THIN SURFACE LAYER, $h_1 \to 0$ OR $\omega \to 0$

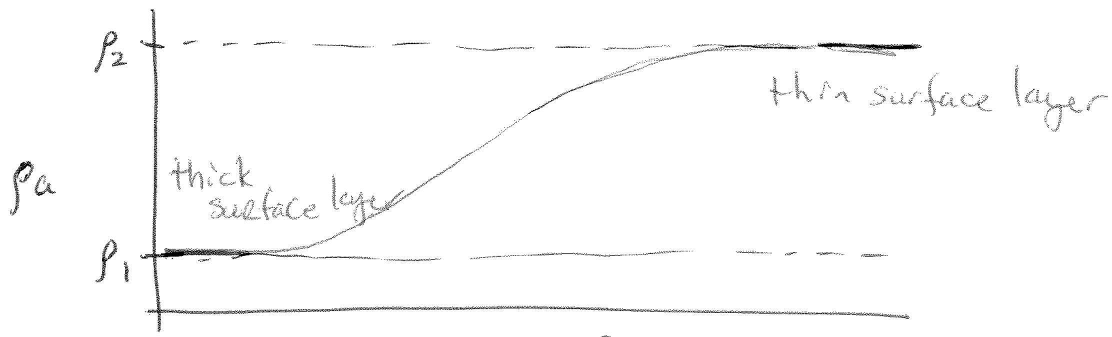
$$Z(0) = \frac{\omega\mu_0}{k_1}\left(\frac{k_1 + k_2 \tanh(ik_1 z)}{k_2 + k_1 \tanh(ik_1 z)}\right)$$

$$\lim_{h_1 \to 0} \tanh(ik_1 h_1) \to 0$$

$$Z(0) = \frac{\omega\mu_0}{\cancel{k_1}}\left(\frac{\cancel{k_1}}{k_2}\right)$$

$$= \frac{\omega\mu_0}{k_2} = Z_2$$

$$\rho_a \to \rho_2$$



$$T = \frac{2\pi}{\omega}$$

HIGH FREQ — LOW FREQ

~~SHORT~~ PERIOD — LONG PERIOD

THIN LAYER APPROXIMATION

$$Z(0) = \frac{\omega\mu_0}{k_1}\left(\frac{k_1 + k_2 \tanh(ik_1 h_1)}{k_2 + k_1 \tanh(ik_1 h_1)}\right)$$

$\lambda = \frac{2\pi}{\omega}$ [m] $\qquad h_1 = [m] \qquad \lambda \gg h_1$

wavelength much bigger

layer thickness

$|ik_1 h_1| \lesssim 1 \qquad \tanh(ik_1 h_1) \approx ik_1 h_1$

$$Z(0) = \omega\mu_0\left(\frac{1 + ik_2 h_1}{ik_1^2 h_1 + k_2}\right)$$